

Network Testing Practical

1. Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other. Hint: On Windows, use 'ping [IP address]' in Command Prompt; on Android, try a ping app or use Termux.

```
C:\Users\Nirali>ping 10.220.185.174

Pinging 10.220.185.174 with 32 bytes of data:
Reply from 10.220.185.174: bytes=32 time=12ms TTL=64
Reply from 10.220.185.174: bytes=32 time=8ms TTL=64
Reply from 10.220.185.174: bytes=32 time=10ms TTL=64
Reply from 10.220.185.174: bytes=32 time=7ms TTL=64

Ping statistics for 10.220.185.174:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 7ms, Maximum = 12ms, Average = 9ms

C:\Users\Nirali>
```

Fig 1. Check that Devices in same network is communicating

2. Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.

```
(root@kali)-[~]
# traceroute www.youtube.com
traceroute to www.youtube.com (142.251.155.4), 30 hops max, 60 byte packets
 1  10.0.2.2 (10.0.2.2)  4.846 ms  4.470 ms  4.129 ms
 2  10.0.2.2 (10.0.2.2)  4.882 ms  4.580 ms  4.424 ms

(root@kali)-[~]
#
```

Fig 2. Traceroute command

3. Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.

```

C:\Users\Nirali>ping 10.220.185.174

Pinging 10.220.185.174 with 32 bytes of data:
General failure.
General failure.
General failure.
General failure.

Ping statistics for 10.220.185.174:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

```

Fig 3. Pinging fail error

4. Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.

```

Interface: 10.220.185.190 --- 0x3
    Internet Address      Physical Address      Type
    10.220.185.174        72-80-bc-6a-e6-50     dynamic
    10.220.185.255        ff-ff-ff-ff-ff-ff     static
    224.0.0.22            01-00-5e-00-00-16     static
    224.0.0.251           01-00-5e-00-00-fb     static
    224.0.0.252           01-00-5e-00-00-fc     static
    239.192.152.143       01-00-5e-40-98-8f     static
    239.255.255.250       01-00-5e-7f-ff-fa     static
    255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 192.168.56.1 --- 0x11
    Internet Address      Physical Address      Type
    192.168.56.255        ff-ff-ff-ff-ff-ff     static
    224.0.0.22            01-00-5e-00-00-16     static
    224.0.0.251           01-00-5e-00-00-fb     static
    224.0.0.252           01-00-5e-00-00-fc     static
    239.192.152.143       01-00-5e-40-98-8f     static
    239.255.255.250       01-00-5e-7f-ff-fa     static

```

Fig 4. List of devices connected to the Wi-Fi

5. Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed. Hint: Include screenshots or command outputs if possible.

1. Objective

- To set up a small network.
- To test network connectivity using **Ping**.
- To check the network path using **Traceroute**.

2. Network Setup

- Connected my **laptop** and **mobile phone** to the same network.
- Checked the laptop's IP address using the **ipconfig** command.
- Verified the IP address, subnet mask, and default gateway.
- Confirmed that the internet connection was working.

3. Ping Test

- Opened **Command Prompt**.
- Ran the **ping** command using the target device's IP address.
- Verified whether the devices could communicate.
- Disconnected the phone from the Wi-Fi network.
- Ran the **ping** command again.
- Received the "**General failure**" error message.

4. Traceroute Test

- Opened the **Terminal** in Kali Linux.
- Ran the command:

tracert www.youtube.com

- Observed the path (hops) from my device to YouTube.
- Counted the total number of hops.

5. Results

- The network was connected successfully.
- The **Ping** command checked communication between devices.
- After disconnecting the phone, **Ping** failed with "**General failure**".
- The **Traceroute** command displayed the route taken by the packets.

6. Issues Observed

- Ping failed after the device was disconnected.
- The error message displayed was "**General failure**".
- The number of hops in **Traceroute** depends on the internet network.

7. Conclusion

- Learned how to set up a basic network.
- Learned how to use **Ping** to test connectivity.
- Learned how to use **Traceroute** to view the network path.
- Understood that disconnected devices cannot communicate on the network.

```

Ethernet adapter Ethernet 3:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::75c9:cc5d:ceb1:25f8%17
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 9:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 10:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 2409:40c1:5e:72a7:7c4f:ed70:184e:9475
    Temporary IPv6 Address. . . . . : 2409:40c1:5e:72a7:f484:432e:7a59:929b
    Link-local IPv6 Address . . . . . : fe80::549a:37c2:ff57:9fb3%3
    IPv4 Address. . . . . : 10.220.185.190
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::7080:bcff:fe6a:e650%3
                                10.220.185.174

```

Fig 5. IP Configuration

```

C:\Users\Nirali>ping 10.220.185.190

Pinging 10.220.185.190 with 32 bytes of data:
Reply from 10.220.185.190: bytes=32 time<1ms TTL=128
Reply from 10.220.185.190: bytes=32 time<1ms TTL=128
|

```

Fig 6. Pinging Device

```

(root@kali)-[~]
# traceroute www.youtube.com
traceroute to www.youtube.com (142.251.155.4), 30 hops max, 60 byte packets
 1  10.0.2.2 (10.0.2.2)  4.846 ms  4.470 ms  4.129 ms
 2  10.0.2.2 (10.0.2.2)  4.882 ms  4.580 ms  4.424 ms

(root@kali)-[~]
#

```

Fig 7. Traceroute

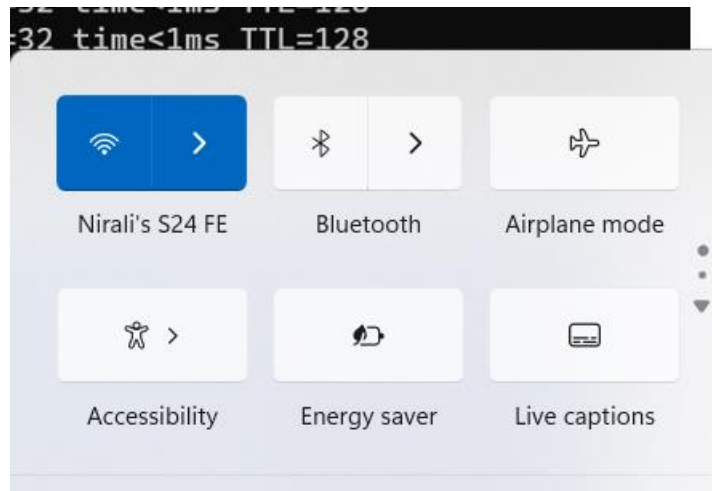


Fig 8. WiFi Connection